

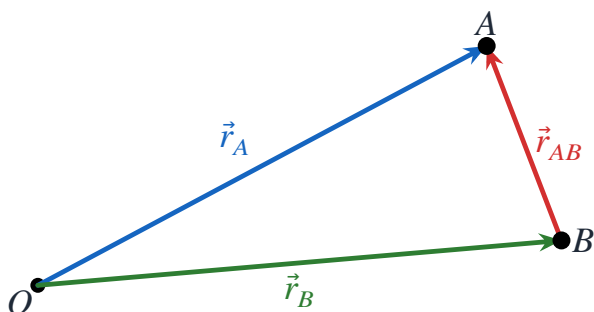
RELATIVE MOTION

Relative Velocity • Along a Line

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1 Relative Position & Velocity

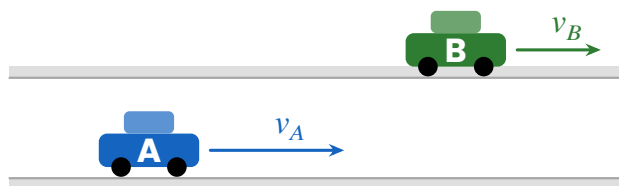
$$\vec{r}_{AB} = \vec{r}_A - \vec{r}_B, \quad \vec{v}_{AB} = \vec{v}_A - \vec{v}_B$$



$\vec{v}_{AB}$: how A moves as seen by B. And
 $\vec{v}_{BA} = -\vec{v}_{AB}$.

2 Same Direction

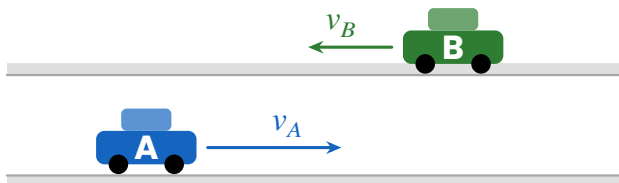
$$v_{AB} = v_A - v_B$$



Overtaking feels slow: only the difference of speeds counts.

3 Opposite Directions

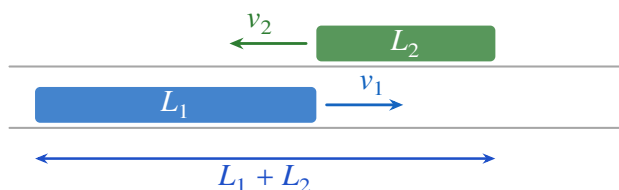
$$v_{AB} = v_A + v_B$$



Head on, the speeds add: the gap closes fast.

4 Crossing Two Trains

$$t = \frac{L_1 + L_2}{v_{\text{rel}}}$$



120 m and 80 m trains at 72 and 54 km/h: opposite 5.71 s, same way 40 s.

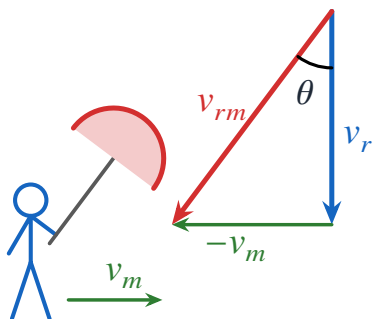
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Rain-Man Problems

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5 Rain in Your Frame

$$\vec{v}_{rm} = \vec{v}_r - \vec{v}_m$$



Walking makes vertical rain come from the front.

6 Umbrella Angle

$$\tan \theta = \frac{v_m}{v_r} = \frac{3}{4} \Rightarrow \theta = 37^\circ$$

$$v_{rm} = \sqrt{v_r^2 + v_m^2} = 5 \text{ m/s}$$

Tilt the umbrella forward by θ from the vertical.

7 Rain Changes With Speed

At 3 km/h the rain looks vertical; at 6 km/h it looks 45° . Find the rain.

$$v_x - 3 = 0, \quad |v_x - 6| = v_y$$

$$\vec{v}_r = (3\hat{i} - 3\hat{j}) \text{ km/h}, \quad 3\sqrt{2} \text{ at } 45^\circ$$

8 Rain With Wind

$$\tan \theta = \frac{v_m - v_w}{v_r}$$

Wind blows the rain along the man's motion; only the horizontal difference tilts it.

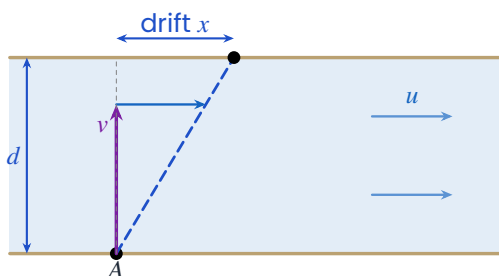
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River-Boat

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9 Shortest Time

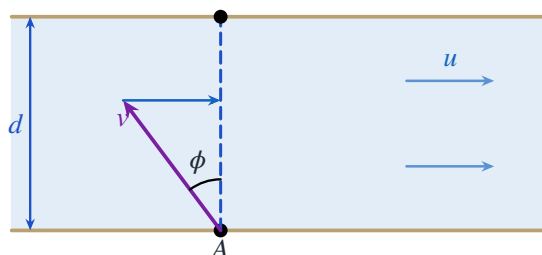
$$t_{\min} = \frac{d}{v}, \quad x = \frac{u d}{v}$$



Head straight across; the river carries you downstream.

10 Zero Drift ($v > u$)

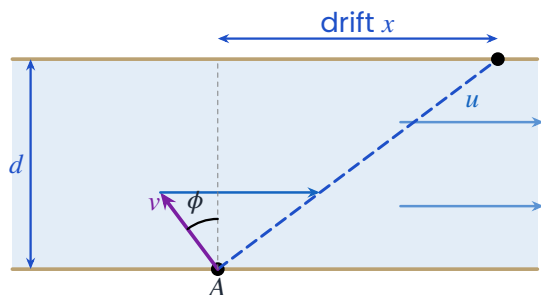
$$\sin \phi = \frac{u}{v}, \quad t = \frac{d}{\sqrt{v^2 - u^2}}$$



Head upstream at ϕ so the ground velocity points straight across.

11 Least Drift ($v < u$)

$$x_{\min} = \frac{d \sqrt{u^2 - v^2}}{v}$$



No zero drift; head at $\sin \phi = v/u$.

12 River Numerical

River 100 m wide flows at 3 m/s; boat speed 5 m/s. Least time? Time with zero drift?

$$t_{\min} = 100/5 = 20 \text{ s}, \quad x = 60 \text{ m}$$

$$\phi = 37^\circ \text{ upstream, } t = 25 \text{ s}$$

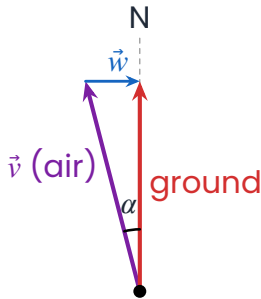
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Wind • Closest Approach • Free Fall

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13 Aeroplane in Wind

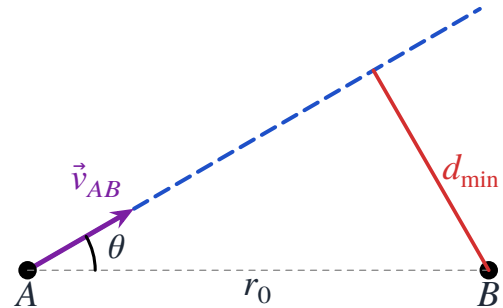
$$\sin \alpha = \frac{w}{v}, \quad v_{\text{ground}} = \sqrt{v^2 - w^2}$$



Head into the wind so its sideways push cancels.

14 Closest Approach

$$d_{\min} = r_0 \sin \theta$$



In B's frame A moves straight along $\vec{v}_{AB}$; drop a perpendicular.

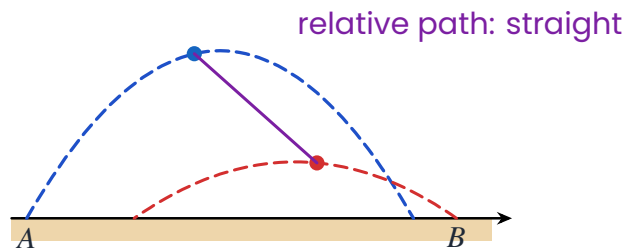
15 When It Happens

$$t = \frac{r_0 \cos \theta}{v_{AB}} = -\frac{\vec{r}_{AB} \cdot \vec{v}_{AB}}{v_{AB}^2}$$

θ is the angle between $-\vec{r}_{AB}$ and $\vec{v}_{AB}$.

16 Free Fall Together

$$\vec{a}_{AB} = \vec{g} - \vec{g} = 0$$



Two projectiles see each other move in a straight line at constant velocity.

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Quick Revision • Traps

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17 Every Case at a Glance

Situation	Result
Same direction / opposite	$v_A - v_B$ / $v_A + v_B$
Crossing or overtaking	$t = (L_1 + L_2) / v_{\text{rel}}$
Rain-man umbrella	$\tan \theta = v_m / v_r$, forward
River, least time	$t = d / v$, drift ud / v
River, zero drift	$\sin \phi = u / v$, $t = d / \sqrt{v^2 - u^2}$
River, $v < u$	$x_{\text{min}} = d \sqrt{u^2 - v^2} / v$
Closest approach	$d_{\text{min}} = r_0 \sin \theta$

18 Common Traps

- $\vec{v}_{AB} = \vec{v}_A - \vec{v}_B$, **never** $\vec{v}_B - \vec{v}_A$.
- Least time and zero drift are different headings.
- In the man's frame the rain's speed grows: $\sqrt{v_r^2 + v_m^2}$.
- Bodies falling together: relative acceleration is zero.

19 Quick Numerical

Car A 72 km/h, car B 54 km/h, same way. Speed of A seen from B?

$$v_{AB} = 20 - 15 = 5 \text{ m/s}$$